

1 4D Rotations Using Matrices

Rotations can also be done using matrix multiplication. We can represent each rotation plane in 4D using a 4×4 matrix.

Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}.$$

1.1 Rotation Matrix Construction

Each rotation matrix is based on the identity matrix in 4D:

$$I_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

A rotation plane changes two axes only. These two axes are represented by a and b .

The values of a and b can be from 0 to 3:

$$0 \leftrightarrow x, \quad 1 \leftrightarrow y, \quad 2 \leftrightarrow z, \quad 3 \leftrightarrow w.$$

We also have the rotation angle θ .

To make the rotation matrix, we start with the identity matrix. Then we change four elements of the 4×4 matrix.

The four elements are

$$\begin{aligned} R_{aa} &= \cos \theta, \\ R_{ab} &= -\sin \theta, \\ R_{ba} &= \sin \theta, \\ R_{bb} &= \cos \theta. \end{aligned}$$

All other elements stay the same as in the identity matrix.

This means that the part of the matrix for the two selected axes looks like the normal 2D rotation matrix:

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The rotated point can then be calculated with

$$\mathbf{x}' = R_{ab}(\theta)\mathbf{x}.$$

1.2 Example

For example, if we choose the x and y axes, then

$$a = 0, \quad b = 1.$$

The rotation matrix becomes

$$R_{xy}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

This rotates the x and y coordinates while leaving z and w unchanged. The new coordinates are

$$\begin{pmatrix} x' \\ y' \\ z' \\ w' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}.$$

This gives

$$\begin{aligned} x' &= x \cos \theta - y \sin \theta, \\ y' &= x \sin \theta + y \cos \theta, \\ z' &= z, \\ w' &= w. \end{aligned}$$

1.3 Possible Rotation Planes

Since we can choose any two different axes, there are six possible rotation planes in 4D:

$$xy, \quad xz, \quad xw, \quad yz, \quad yw, \quad zw.$$

The six planes come from choosing two axes out of four:

$$\binom{4}{2} = 6.$$

The same method can be used for all six planes. We start with the identity matrix and change the four elements belonging to the selected axes.

1.4 Implementation

In the code, the two selected axes are stored as **axis_a** and **axis_b**.

The four elements of the matrix are changed as follows:

$$\begin{aligned} R[\text{axis_a}][\text{axis_a}] &\leftarrow \cos \theta, \\ R[\text{axis_a}][\text{axis_b}] &\leftarrow -\sin \theta, \\ R[\text{axis_b}][\text{axis_a}] &\leftarrow \sin \theta, \\ R[\text{axis_b}][\text{axis_b}] &\leftarrow \cos \theta. \end{aligned}$$

After these changes, the matrix represents the rotation for the selected pair of axes.